CSE 270 Sports Analytics

Predictions based on Performance Ratings

Narek Sahakyan, Habet Madoyan

Predictions based on Performance Ratings

Libraries used

```
library(dplyr)
library(elo)
library(ggplot2)
library(SportsAnalytics270)
library(caret)
```

Predictions based on Performance Ratings

Elo ratings

A rating is a memory

- A rating is not a trophy. It is a memory of how surprising you have been.
- **The update.** You gain points for being better than we thought, not for being good.
- **The league.** We will ask whether the Premier League ranking still sounds like football.

Elo rating system

- Is initially developed by Hungarian-american professor of physics Arpad Elo.
- Initially was used to rank chess players.
- Can be used to predict the game outcome for any two-players or two-teams game.



Arpad Elo

Elo rating system

- Assign performance ratings to each team/player. This rating is going to show the ability of the team to win.
- After each game update the the player's rating based on the outcome.
- Use the ability scores to predict the match outcome of the game

Elo rating system

The general formula

The expected probability of winning a game by Team A will be calculated by the following formula.

For home team

$$Y^H = \frac{1}{1+c^{(R^A-\theta-R^H)/d}}$$

For away team

$$Y^A = 1 - \frac{1}{1+c^{(R^A-\theta-R^H)/d}}$$

Where R^H, R^A are home and away teams Ratings, θ is some context parameter (can be omitted) such as home team advantage.
The parameters c and d are used to scale the ratings and are usually taken as c=10, d=400

Elo rating system

- Why are c=10, d=400.
- This is just for convention and scaling. If Player A is 10 times more likely to beat Player B (Odds), then the difference in the Elo scores needs to be 400.
- Assume teams playing in a neutral field $\theta = 0$

$$Y^H = \frac{1}{1+10^{(R^A-R^H)/400}}, Y^A = 1 - \frac{1}{1+10^{(R^A-R^H)/400}}$$

$$Odds(\text{H beats A}) = \frac{Y^H}{Y^A} = \frac{\frac{1}{1+10^{(R^A-R^H)/400}}}{1-\frac{1}{1+10^{(R^A-R^H)/400}}} = \frac{\frac{1}{1+10^{(R^A-R^H)/400}}}{\frac{10^{(R^A-R^H)/400}}{1+10^{(R^A-R^H)/400}}} = 10^{(R^H-R^A)/400}$$

If $R^H - R^A = 400$, then $Odds(\text{H beats A}) = 10$

Elo rating system

- Several R packages for elo rating calculations
- We will use mostly “elo”

```
install.packages("PlayerRatings")
install.packages("elo")
install.packages("EloRating")
```

Before we calculate

- Two teams, both rated 1500. Who wins?
- 1900 versus 1500. Hands for the 1900.

Elo rating system

- predict probability to win for player A and player B
- Calculate the odds to win

```
elo.prob(elo.A=1500, elo.B=1500)
```

```
## [1] 0.5
```

```
elo.prob(elo.A=1900, elo.B=1500)
```

```
## [1] 0.9090909
```

Elo rating system

c = 10, d = 400 parameters

```
elo.prob(elo.A=1900, elo.B=1500)
```

```
## [1] 0.9090909
```

```
# The odds
elo.prob(elo.A=1900, elo.B=1500)/(1-elo.prob(elo.A=1900, elo.B=1500))
```

```
## [1] 10
```

With the difference of 400 Elo rating points, it is 10 times more likely that Team A will win

The scale

- That fraction is the odds.
- The scale is built so that 400 points equals 10-to-1.

Elo rating system

The rating is updated with the following formula

$R_T^H = R_{t-1}^H + K * (\alpha^H - y^H)$

where

R_t^H updated rating for the home team

R_{t-1}^H rating before the game

α^H is the actual outcome for the game

y^H is the predicted probability of Home team to win

K is a positive number, which controls how drastically the rating is changed

Elo rating system

The actual score for the home team α^H is given by

$$\alpha^H = \begin{cases} 1, \textit{HomeTeamWon} \\ 0.5, \textit{Draw} \\ 0, \textit{Otherwise} \end{cases}$$

In football/soccer the coefficient K can be updated based on the goal differential

$$K = k_0(1+\delta)^{\lambda}$$

Where λ is the absolute goal difference, k_0, λ are positive constants There are several suggestions for the value of K, we will take it equal to 20.

Before the update

- Player A is 1500. Player B is 1600. A wins.
- Who goes up? By a lot or a little?
- What if they had drawn instead?
- A draw hurts the favorite, because 0.5 was not what the rating expected.

Elo rating system

example: Two players

- Player A, Elo rating of 1500
- Player B, Elo rating of 1600

$$R_T^H = R_{t-1}^H + K * (\alpha^H - y^H)$$

```
# prob for A
elo.prob(elo.A=1500, elo.B=1600)
```

```
## [1] 0.359935
```

```
1500+20*(1-0.359935)
```

```
## [1] 1512.801
```

```
elo.prob(elo.A=1600, elo.B=1500)
```

```
## [1] 0.640065
```

```
1600+20*(0-0.640065)
```

```
## [1] 1587.199
```

Elo rating system

- The player A will get an increase in rating by 12.8013
- The player B will get a decrease in rating by 12.8013
- 12.8013 is calculated using function elo.update()

```
elo.calc(elo.A=1500, elo.B=1600, wins.A=1, k = 20)
```

```
##      elo.A    elo.B
## 1 1512.801 1587.199
```

```
elo.update(elo.A=1500,elo.B=1600,wins.A=1, k=20)
```

```
## [1] 12.8013
```

Elo rating system

Dataset on EPL games for seasons 2000-2026

```
data("epl")
str(epl)
```

```
## 'data.frame':    9880 obs. of  6 variables:
##  $ Home  : chr  "Charlton Athletic" "Chelsea FC" "Coventry City" "Derby County" .
##  $ Away  : chr  "Manchester City" "West Ham United" "Middlesbrough FC" "Southampt
##  $ HG    : num  4 4 1 2 2 0 1 1 3 2 ...
##  $ AG    : num  0 2 3 2 0 0 0 0 1 0 ...
##  $ Round : num  1 1 1 1 1 1 1 1 1 1 ...
##  $ Season: num  2001 2001 2001 2001 2001 ...
```

Elo rating system

score function from Elo package does a very simple calculation: Based on the game score returns a vector with the results, 1 if team A won, 0.5 if it was a draw and 0
if Team A lost

3 games, 2-0, 0-0, 0-1

```
score(2,0)
```

```
## [1] 1
```

```
score(0,0)
```

```
## [1] 0.5
```

```
score(0,1)
```

```
## [1] 0
```

Elo rating system

Run Elo model

```
elos <- elo.run(score(HG, AG) ~ Home + Away,
        data = epl, k = 20)
elos_df <- as.data.frame(elos)
```

score(HG, AG) calculates the final score of the game based on Home Goal and Away Goal statistics

Elo rating system: Choosing K

- K values can be updated based on the importance of the games
- The Eloratings.net uses the following values for k for national teams

Competition	Multiplier
Friendly match	20
All minor tournaments	30
World Cup and continental qualifiers and major tournaments	40
Continental championship finals and major intercontinental tournaments	50
FIFA World Cup finals	60

Elo rating system

- Look at the dataframe
- initial values for Elo are set to 1500

```
head(elos_df)
```

```
##                team.A           team.B p.A wins.A update.A update.B elo.A elo.B
## 1 Charlton Athletic  Manchester City 0.5    1.0       10      -10  1510  1490
## 2        Chelsea FC  West Ham United 0.5    1.0       10      -10  1510  1490
## 3     Coventry City Middlesbrough FC 0.5    0.0      -10       10  1490  1510
## 4      Derby County   Southampton FC 0.5    0.5        0        0  1500  1500
## 5      Leeds United       Everton FC 0.5    1.0       10      -10  1510  1490
## 6    Leicester City      Aston Villa 0.5    0.5        0        0  1500  1500
```

Elo rating system

- The final elo ratings
- n=10 as there are 10 games each round

```
tail(elos_df, n=10)
```

```
##                     team.A                 team.B       p.A wins.A   update.A
## 9871         Sunderland AFC             Chelsea FC 0.3758739    1.0  12.482522
## 9872 Brighton & Hove Albion      Manchester United 0.4293548    0.0  -8.587096
## 9873         Crystal Palace            Arsenal FC 0.2257292    0.0  -4.514584
## 9874             Burnley FC Wolverhampton Wanderers 0.4282338    0.5   1.435324
## 9875              Fulham FC       Newcastle United 0.4500734    1.0  10.998533
## 9876           Liverpool FC           Brentford FC 0.6179266    0.5  -2.358531
## 9877        Manchester City            Aston Villa 0.6833066    0.0 -13.666132
## 9878      Nottingham Forest        AFC Bournemouth 0.4010271    0.5   1.979459
## 9879      Tottenham Hotspur             Everton FC 0.3920601    1.0  12.158798
## 9880        West Ham United           Leeds United 0.4499209    1.0  11.001581
##        update.B    elo.A    elo.B
## 9871 -12.482522 1546.127 1609.254
## 9872   8.587096 1612.340 1678.934
## 9873   4.514584 1569.241 1792.393
## 9874  -1.435324 1407.752 1455.096
## 9875 -10.998533 1584.474 1597.286
## 9876   2.358531 1682.986 1604.187
## 9877  13.666132 1771.818 1665.560
## 9878  -1.979459 1579.851 1645.586
## 9879 -12.158798 1520.586 1572.471
## 9880 -11.001581 1536.481 1549.394
```

Before we sort

- Shout a club you think finishes number one after twenty-five years of the Premier League.
- If the room is wrong, that is the point.
- The rating remembers results, not reputation.

Elo rating system

Get the final elos

```
elo_final <- final.elos(elos)
sort(elo_final, decreasing = T)
```

```
##              Arsenal FC         Manchester City            Liverpool FC
##                1792.393                1771.818                1682.986
##       Manchester United             Aston Villa         AFC Bournemouth
##                1678.934                1665.560                1645.586
##  Brighton & Hove Albion              Chelsea FC            Brentford FC
##                1612.340                1609.254                1604.187
##        Newcastle United              Fulham FC       Nottingham Forest
##                1597.286                1584.474                1579.851
##              Everton FC          Crystal Palace            Leeds United
##                1572.471                1569.241                1549.394
##          Sunderland AFC         West Ham United       Tottenham Hotspur
##                1546.127                1536.481                1520.586
##          Wigan Athletic              Stoke City         Birmingham City
##                1488.608                1484.236                1483.599
##            Swansea City            Blackpool FC        Bolton Wanderers
##                1473.755                1471.149                1467.713
##       Charlton Athletic Wolverhampton Wanderers        Blackburn Rovers
##                1458.265                1455.096                1447.924
##              Hull City              Luton Town          Leicester City
##                1441.610                1439.436                1438.098
##            Cardiff City     West Bromwich Albion           Portsmouth FC
##                1434.696                1433.524                1431.484
##          Coventry City              Reading FC        Middlesbrough FC
##                1428.123                1423.624                1415.869
##            Ipswich Town              Burnley FC     Queens Park Rangers
##                1409.904                1407.752                1405.656
##              Watford FC           Bradford City       Huddersfield Town
##                1397.591                1397.343                1371.415
##        Sheffield United            Norwich City          Southampton FC
##                1366.805                1362.177                1350.971
##             Derby County
```

How big is a home crowd?

- How many Elo points is a home crowd worth? 50, 200, or 400?
- 200 is half of the famous 400.
- Equal teams: the home side starts as a moderate favorite, before a ball is kicked.

Elo rating system

- Taking into account Home advantage
- θ Home advantage parameter $= 200$
- This means that for every game, 200 is added to the home teams Elo rating

$$Y^H = \frac{1}{1+c^{(R^A - 200 - R^H)/d}}$$

$$Y^A = 1 - \frac{1}{1+c^{(R^A - 200 - R^H)/d}}$$

Elo rating system

```
epl2 <- elo.run(score(HG, AG) ~ adjust(Home, 200) +
                                Away,  data = epl, k = 20)
```

```
sort(final.elos(epl2), decreasing = T)
```

```
##              Arsenal FC         Manchester City            Liverpool FC
##                1858.811                1844.120                1750.513
##       Manchester United             Aston Villa          AFC Bournemouth
##                1708.087                1705.865                1658.729
##              Chelsea FC  Brighton & Hove Albion         Newcastle United
##                1649.302                1637.592                1625.674
##            Brentford FC               Fulham FC              Everton FC
##                1619.713                1592.715                1587.358
##       Nottingham Forest          Crystal Palace            Leeds United
##                1587.199                1586.457                1544.102
##         West Ham United          Sunderland AFC       Tottenham Hotspur
##                1536.880                1531.760                1531.284
##              Stoke City          Wigan Athletic         Birmingham City
##                1484.091                1478.457                1476.486
##            Swansea City            Blackpool FC         Bolton Wanderers
##                1467.436                1460.175                1450.821
##       Charlton Athletic Wolverhampton Wanderers         Blackburn Rovers
##                1443.971                1442.937                1440.777
##          Leicester City              Luton Town           Portsmouth FC
##                1435.375                1426.773                1421.569
##             Cardiff City           Coventry City               Hull City
##                1419.298                1417.951                1416.978
##     West Bromwich Albion             Reading FC         Middlesbrough FC
##                1416.249                1407.307                1398.166
##             Ipswich Town     Queens Park Rangers            Bradford City
##                1396.594                1384.800                1384.045
##               Burnley FC             Watford FC         Huddersfield Town
##                1382.679                1377.189                1354.533
##         Sheffield United           Norwich City           Southampton FC
##                1341.319                1328.660                1327.539
```

Does the margin matter?

- Should winning 4–0 move the rating more than winning 1–0?

Elo rating system

Defining dynamic K factor

Update the Elo rating formula using the following equation

$$R_t^H = R_{t-1}^H + K * G * (\alpha^H - y^H)$$

Where G is a goal difference factor. The website Eloratings.net uses the following formula for G Factor.

$$G = \begin{cases} 1, N <= 1 \\ 1.5, N = 2 \\ (N + 11)/8, N >= 3 \end{cases}$$

Elo rating system

Adding the goal difference factor

```
epl$G <- ifelse(abs(epl$HG-epl$AG)<=1, 1,
                    ifelse(abs(epl$HG-epl$AG)==2,1.5,
                           (abs(epl$HG-epl$AG)+11)/8))
```

```
head(epl,5)
```

```
##                 Home              Away HG AG Round Season     G
## 1 Charlton Athletic   Manchester City  4  0     1   2001 1.875
## 2        Chelsea FC   West Ham United  4  2     1   2001 1.500
## 3     Coventry City Middlesbrough FC  1  3     1   2001 1.500
## 4      Derby County    Southampton FC  2  2     1   2001 1.000
## 5      Leeds United        Everton FC  2  0     1   2001 1.500
```

Elo rating system

```
epl3<-elo.run(score(HG, AG) ~ adjust(Home, 200) + Away +
                              k(20*epl$G),  data = epl )
```

```
sort(final.elos(epl3), decreasing = T)
```

```
##             Arsenal FC        Manchester City           Liverpool FC
##               1914.509               1896.792               1787.608
##       Manchester United            Aston Villa         AFC Bournemouth
##               1738.978               1717.973               1697.510
##             Chelsea FC  Brighton & Hove Albion        Newcastle United
##               1674.871               1668.327               1656.074
##           Brentford FC       Nottingham Forest          Crystal Palace
##               1649.045               1638.069               1616.266
##              Fulham FC          Everton FC             Leeds United
##               1615.722               1596.389               1556.899
##         West Ham United       Tottenham Hotspur          Sunderland AFC
##               1550.052               1549.650               1532.393
##          Wigan Athletic           Stoke City            Swansea City
##               1479.286               1476.806               1468.772
##         Birmingham City        Blackpool FC         Blackburn Rovers
##               1454.399               1452.081               1440.332
##        Bolton Wanderers Wolverhampton Wanderers       Charlton Athletic
##               1436.024               1433.375               1424.656
##           Portsmouth FC          Luton Town              Hull City
##               1420.757               1413.331               1410.687
##          Leicester City          Cardiff City           Coventry City
##               1403.533               1400.593               1396.630
##     Queens Park Rangers    West Bromwich Albion            Reading FC
##               1395.745               1392.923               1388.352
##        Middlesbrough FC          Ipswich Town             Burnley FC
##               1388.068               1368.077               1367.834
##           Bradford City            Watford FC       Huddersfield Town
##               1353.378               1349.938               1329.647
##        Sheffield United          Norwich City          Southampton FC
##               1295.796               1292.760               1288.715
```

Before the accuracy

- What accuracy would make you bet with this?
- What accuracy would make you shrug?
- The next fit uses a home adjustment of 50, not 200. The crowd just got smaller.
- The band from 0.45 to 0.55 is Elo saying “I do not know.” A predicted draw is humility, not a bug.

Elo rating system

- Can we predict the actual outcome and not just the probability ?
- Result for the home team

$$\text{Predicted outcome} = \begin{cases} 1, p >= 0.55 \\ 0.5, 0.45 < p < 0.55 \\ 0, p <= 0,45 \end{cases}$$

```
epl2 <- elo.run(score(HG, AG) ~ adjust(Home, 50) +
                                Away,  data = epl, k = 20)
epl2 <- as.data.frame(epl2)

epl2$Predicted <- ifelse(epl2$p.A>=0.55, 1,
                         ifelse(epl2$p.A<=0.45,0,0.5))
library(caret)
confusion <- table(epl2$wins.A, epl2$Predicted)
```

```
sum(diag(confusion))/nrow(epl2)
```

```
## [1] 0.5120445
```

End of the season

Does the top team from the previous season always perform as good in the next season ?

Elo rating system

- The teams change every season, players come and go
- The rating of the previous season cannot be carried fully to the new season
- The following formula is suggested by the MARS rating system

New Season Starting Rating = 1500+(Previous Season Final Rating-1500)/2

For newly promoted teams

New Season Starting Rating = 1500+(Average Final Rating of relegated teams-1500)/2)

Elo rating system

To get the Elo rating working better

- Tune home advantage parameter
- Tune new teams initial Elo's parameters
- different K parameters for different games
- Think of what else can influence the game

What else should count?

- What else should move a rating besides the score?
- Injuries, the schedule, a derby.
- A rating is not a trophy. It is a memory of how surprising you have been.

Elo rating system

Lot of different web-sites provides elo ratings in different sports

- Some mix of elo ratings and optimizations from FiveThirtyEight on NBA teams and games https://projects.fivethirtyeight.com/carmelo/
- Elo ratings from nfl https://projects.fivethirtyeight.com/complete-history-of-the-nfl/
- Football team elos
- European clubs elo ratings http://clubelo.com/
- for national teams http://eloratings.net

In-class assignment

TIME FOR IN CLASS ACTIVITY !

A rating moves when the result is a surprise. This takes more than five minutes. Work in pairs.

Run the same three Elo models on **three slices** of `epl`. The table starts in 2001. Fit each slice from scratch.

- 2000–2010
- 2010–2020
- 2020–2026

On each slice:

- $K = 20$, no home adjustment
- $K = 20$, home adjustment 200
- home adjustment 200, and the goal-difference factor G from the lecture
- Pick one club. From the **second** model, find three of its games as team A: a win, a draw, and a loss. Write down `p.A`, `wins.A`, and `update.A`.

In-class assignment

Answer these for each slice.

- In which game was the result most surprising?
- When the favorite won, did the rating move a lot or a little?
- Did the top 5 change between the first model and the third?